John Ian R. So

johnianrso@berkeley.edu johnianrso johnrso (925) 336-7553

Education

University of California, Berkeley **GPA: 4.00 / 4.00**

B.S. Electrical Engineering and Computer Science, May 2023

Courses Completed: Structure & Interpretation of Computer Programs, Data Structures, Machine Structures, Discrete Mathematics & Probability Theory, Designing Information Devices & Systems I, Self-Driving Cars, Multivariable Calculus, Physics for Scientists & Engineers.

Courses in Progress: Efficient Algorithms & Intractable Problems, Designing Information Devices & Systems II.

Technical Experience

Robotics Intern, PhaseSpace Motion Capture *May 2020-August 2020*

- Researched high contact surface sanitization methods, including UV-C and hypochlorous acid (HOCl), as a response to the COVID-19 pandemic.
- Prototyped software, firmware, and hardware using an ESP-32 controller and Firebase for a web interfaced, closed loop control system for a HOCl generator and dispenser system.
- Wrote and submitted a project proposal to the Department of Homeland Security SVIP Program for $200,000 for the HOCl system. Results pending.
- Implemented robot system architecture in ROS Melodic. Modelled a robot frame using URDF and simulated sensors and actuators in Gazebo.
- Designed a navigation software stack based on Real-Time Appearance-Based Mapping, Adaptive Monte Carlo Localization, and visual and IMU odometry.

Lead Intern, PhaseSpace Motion Capture *May 2018-August 2018*

- Led a team of four in building a robotic arm driven by an ESP-32 controller and PhaseSpace's proprietary motion capture system.
- Created a control system in Arduino to translate 3D vectors to arm joint angles.
- Developed software in Java to translate motion capture data into scaled vector representations and feed the data to the arm's control system.
- Machined the arm from scratch using parts modelled in Autodesk Fusion 360.

EECS Engineer, Formula Electric at Berkeley *January 2020-Present*

- Engineered designs for the tractive system indicator and driver interface subsystems of an electric vehicle for the Formula SAE Competition EV division.
- Completed PCB designs for the tractive system indicator and driver interface in Altium Designer. Currently creating firmware for STM controllers in Mbed OS.

Teaching Experience

Course Staff, CS 61A *August 2020-Present*

- Leading tutorials for discussions of five students four times per week for course content, including Python and programming fundamentals.
- Holding drop-in office hours twice per week to reinforce course concepts.
- Developing software for internal staff tools, including Slack bot integration.

Mentor, Computer Science Mentors *January 2020-Present*

- Leading weekly tutoring sessions for students in CS 61A, covering functional, object-oriented, and declarative programming in Python, Scheme, and SQL.

Skills

Languages: Java, C, C++, JavaScript, Python, Ruby, RISC-V, SQL, HTML/CSS.
Packages/Frameworks: ReactJS, Flask, Android Studio, OpenMP.
Robotics: ROS, OpenCV, Keras, TensorFlow, Arduino, ESP.
Tools: Git, Unix Shell, IntelliJ.
Circuit/PCB Design: Altium Designer, Multisim, Autodesk Eagle.

Projects

Mask Vision *August 2020-Present*

- Researched and implemented CNNs using Keras for detection of masks in images.
- Automated preparation of data sets using OpenCV and Keras. Currently looking into transfer learning using the YOLO architecture for real-time recognition of proper and improper usage of masks.

Word Hunt Solver *May 2020*

- Built a model-view-controller program for Word Hunt, a game where players swipe words on a 4x4 board to earn points.
- Implemented a solver that finds all possible words on the board within one second using DFS graph traversal.
- Currently deploying as a web application using Heroku, Flask, and ReactJS.

Mixer, CalHacks 6.0 *October 2019*

- Wrote a program using Spotify REST APIs that generates song recommendations based on seed songs and genres by scraping public playlists.
- Designed an algorithm to find similar songs and uploaded the selected songs to a Spotify playlist. Uploaded results to Firebase for future use.

Self-Driving Car Simulation *December 2019*

- Modelled and implemented self-driving car subsystems such as acceleration control, route planning, and steering control in Python using the Duckietown environment, OpenCV, and TensorFlow.
- Created models for linear regression and neural networks. Used PID, A*, and cost maps to plan and run routes.

NumC *July 2020-August 2020*

- Implemented a matrix library in C and a Python-C interface for use in Python. Supports element-wise addition, subtraction, negation, absolute value, and matrix multiplication/exponentiation.
- Optimized matrix operations using Intel SSE Intrinsics, OpenMP, loop unrolling, and cache blocking. Achieved a ~2400x speedup compared to unoptimized benchmarks.

Interests

Robotic perception & manipulation, autonomous vehicles, parallel computing, game theory, small scale computing